Summary Sheet

Ravi Bajaj | AI in Healthcare | Final paper: A Deterministic Evidence Gate for Pharmacogenomic LLM Alerts: Synthetic Stress-Testing with Ablation

Conference / Symposium

Field	Value
Name	Machine Learning for Health Symposium (ML4H 2026)
Conference URL	https://ml4h.ahli.cc/
Instructions / CFP	https://ml4h.ahli.cc/submit/call-for-papers/
Submission deadline	September 10, 2026, 11:59 PM AoE
Author notification	October 22, 2026
Camera-ready deadline	November 7, 2026 (tentative)
Event dates	December 6-7, 2026
Event location	Sydney, Australia
Suggested track	Findings Track
Editable paper format	LaTeX/Overleaf source: paper/ml4h_findings_evidence_gate.tex and paper/ml4h_findings_refs.bib

Final Paper Title and Abstract

Title: A Deterministic Evidence Gate for Pharmacogenomic LLM Alerts: Synthetic Stress-Testing with Ablation

Abstract: LLM-generated medication alerts can sound more certain than their evidence permits. This capstone tests a deterministic evidence gate for pharmacogenomic alert drafts using 33 author-designed synthetic cases: 23 overclaim archetypes and 10 bounded aligned alerts. The gate consumes structured annotations for source support, population fit, endpoint strength, and actionability, then returns allow, narrow, abstain, or deny. With oracle annotations, the full monitor routed 23/23 overclaims and preserved 10/10 bounded alerts. With citation fields, GPT-5.6-terra and Grok-4.5 each matched 21/33 downstream actions, while preserving 4/10 and 9/10 bounded alerts. Without citation fields, downstream action agreement was 11/33 and 12/33, with 0/10 and 2/10 bounded alerts preserved. Disabling source support, population fit, or claim strength let 2/23, 1/23, and 6/23 overclaims pass unchanged. These are specification-conformance and extraction-bottleneck results on synthetic cases, not clinical safety, generalization, or patient-benefit evidence.

Generative AI Use Disclosure

Generative AI tools were used for brainstorming, critique, code scaffolding, drafting, formatting, and synthetic LLM experiments. GPT-5.6-terra and Grok-4.5 were used in structured-label translation and text-only extraction experiments; GPT-5.6-sol generated unlabeled held-out scenarios for later author labeling. No real patient records, protected health information, or private genomic data were used or transmitted. The author selected the final scope, verified sources, ran the code, reviewed outputs, and remains responsible for all claims and limitations.

Optional Data, Code, and Tools

- Synthetic 33-case pharmacogenomic/genomic medication-alert case bank.
- Deterministic three-check monitor with unit tests and replayable CSV/JSON outputs.
- Ablation, policy-comparator, structured-label translation, text-only extraction, derived directional-error/PRF, and held-out authoring artifacts under code/ and results/.
- Safety boundary: no diagnosis, no treatment recommendation, no real patient data, and no protected health information.